LEVELLING

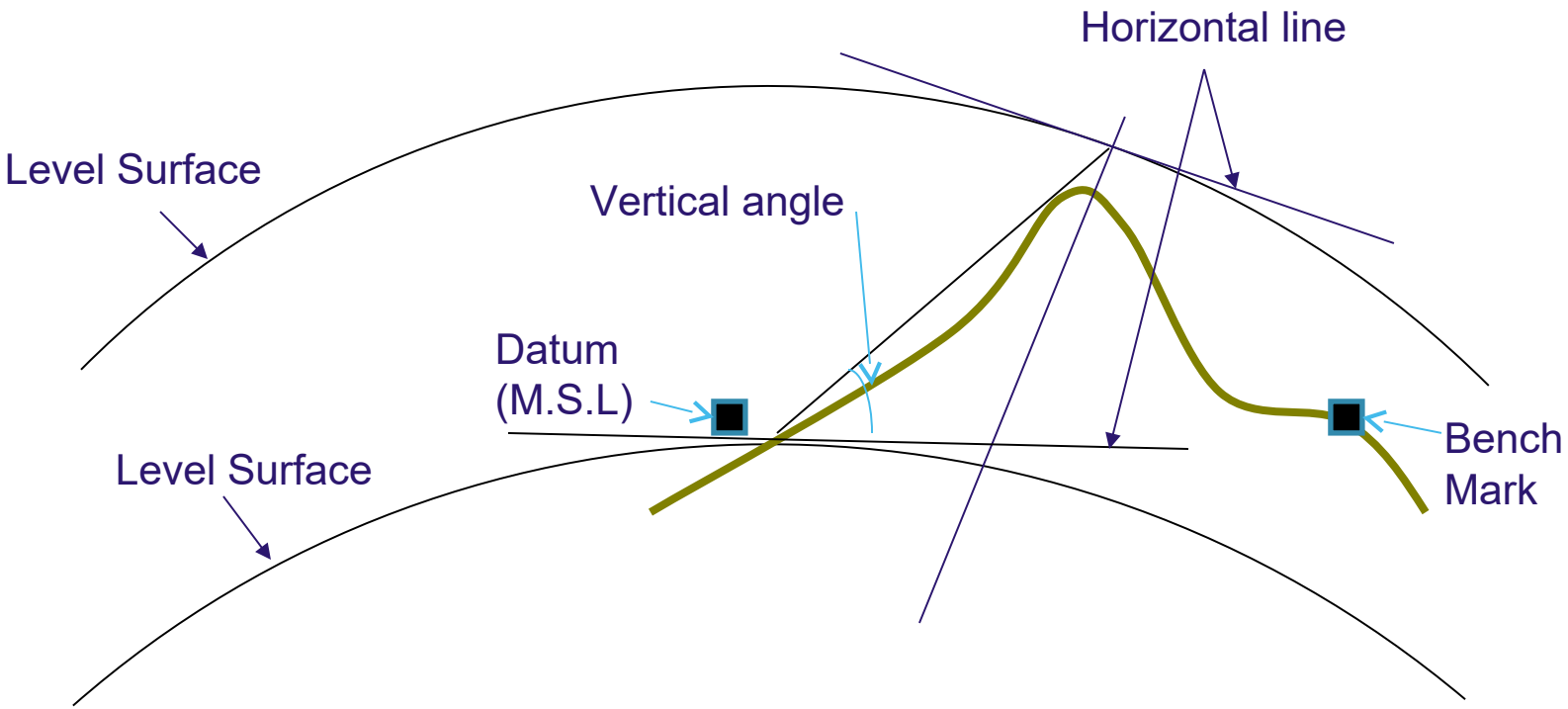
INTRODUCTION & BASIC DEFINITIONS

LEVELLING:

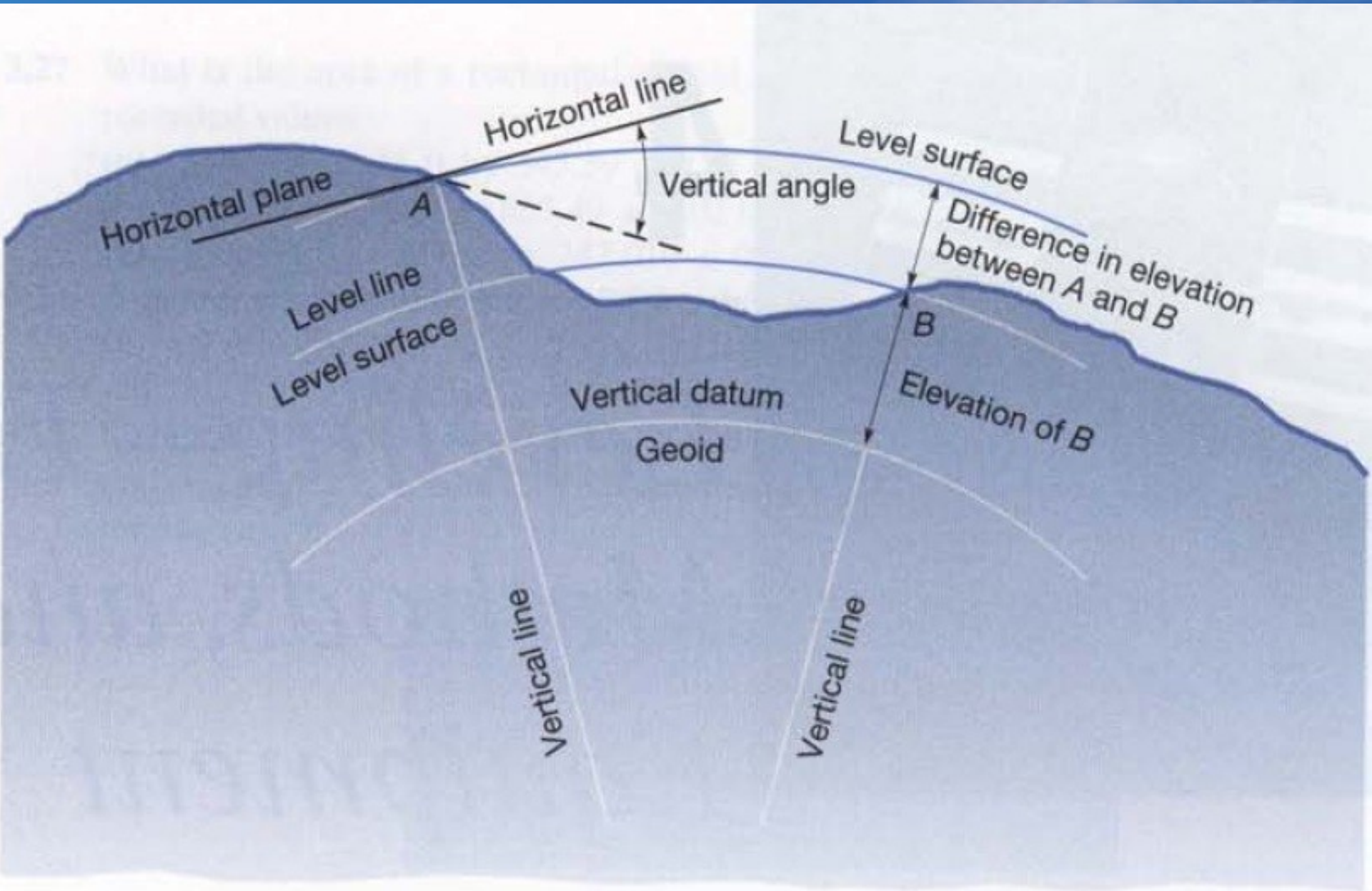
Leveling may be defined as the art of determining the relative heights or elevations of points or objects on the earth's surface. It deals with measurements in a vertical plane.

Definitions of Terms used in Leveling: -

- **A level surface** is any surface parallel to the mean spheroidal surface of the earth. A level surface may be regarded as a curved surface, every point on which is equidistant from the centre of the earth. It is normal to the plumb line at all points.
- **A level line** is a line lying in a level surface. It is, therefore, normal to the plumb line at all points


Horizontal line
Level Surface
Vertical angle
Datum
(M.S.L)
Level Surface
Bench
Mark

INTRODUCTION & BASIC DEFINITIONS



INTRODUCTION & BASIC DEFINITIONS

- **A datum surface or line** is any arbitrarily assumed level surface or line from which vertical distances are measured.
- **The elevation of a point** is its vertical distance above or below the datum. It is also known as the reduced level (R.L). The elevation of a point is plus or minus according as the point as above or below the datum.
- **The difference in elevation (H)** between two points is the vertical distance between the level surfaces passing through the two points.
- **A bench mark (B.M)** is a fixed reference point of known elevation.

INTRODUCTION & BASIC DEFINITIONS

- **The line of collimation** is the line joining the intersection of the crosshairs to the optical centre of the object glass and its continuation. It is also called the line of sight.
- **A backsight (B.S),** also termed as a backsight reading, is a staff (or rod) reading taken on a point of known elevation, as on a bench mark or a change point. It is also called a plus sight. It is the first staff reading taken after the level is set up and leveled.
- **A foresight (F.S) also called a foresight reading,** is a staff (or rod) reading on a point whose elevation is to be determined as on a change point. It is also termed as a minus sight. It is the last staff reading denoting the shifting of the instrument.

INTRODUCTION & BASIC DEFINITIONS

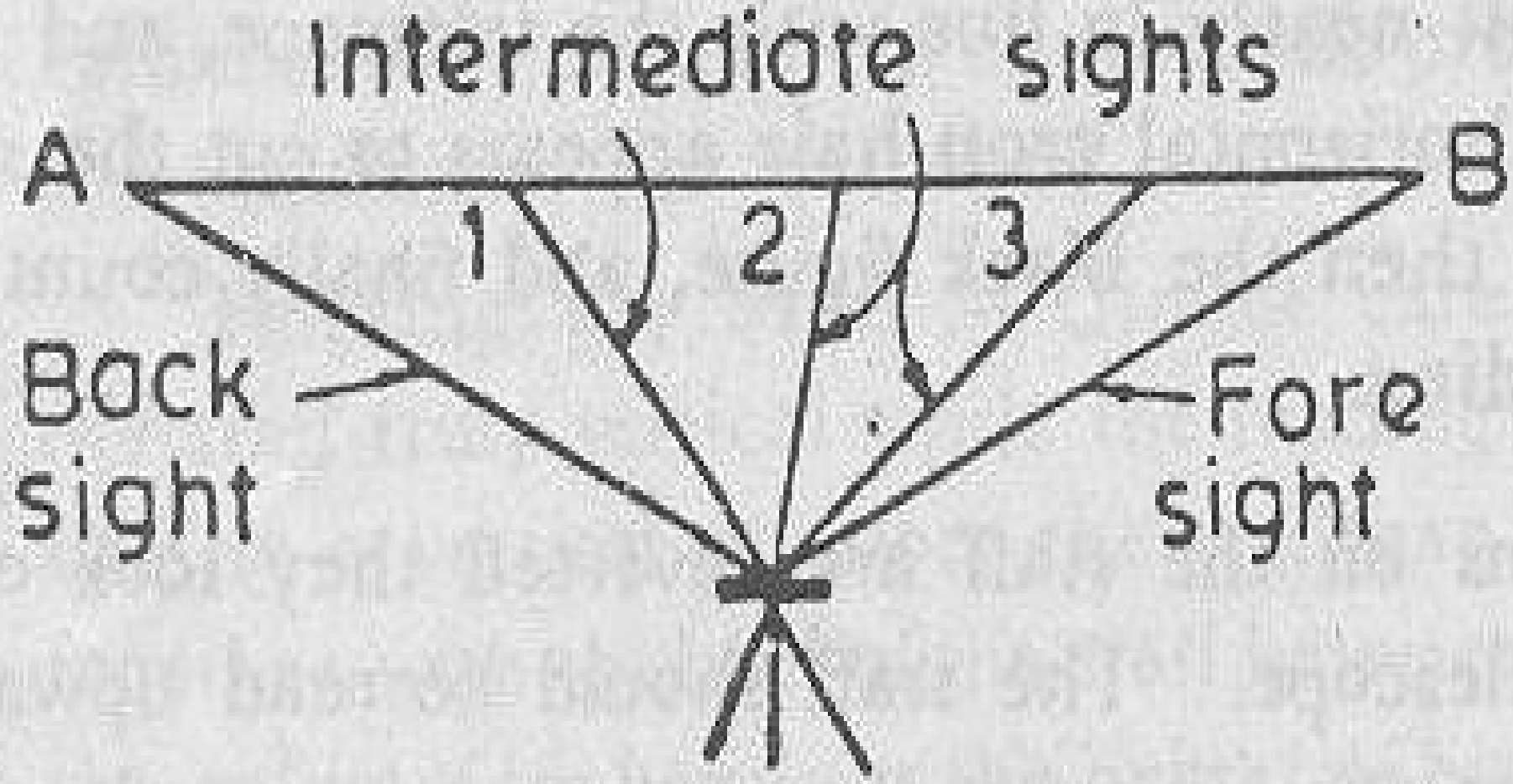


Fig. 255

INTRODUCTION & BASIC DEFINITIONS

- **An intermediate sight (I.S)** is any other staff reading taken on a point of unknown elevation from the same set-up of the level. All sights taken between the backsight and the foresight are intermediate sights.

- **A change point (C.P)** is a point denoting the shifting of the level. It is a point on which the fore and backsight are taken. Any stable and well-defined object such as a boundary stone, curb stone, rail, rock, etc. is used as a change point. A bench mark may also be taken as a change point. It is also called a turning point (T.P).

INTRODUCTION & BASIC DEFINITIONS

- **A station** is a point whose elevation is to be determined or a point which is to be established at a given elevation. It may be noted that it is a point where the staff is held and not the point where the level is set up.
- **The height of instrument (H.I)** is the elevation (or the R.L) of the plane of collimation (or plane of sight) when the instrument is correctly leveled. It is also called the "height of plane of collimation" or the collimation.

EQUIPMENT FOR LEVELING

The equipment for leveling consists of

- (i) Engineering Level,
- (ii) Leveling staff,
- (iii) Chain or tape for measuring horizontal distances
- (iv) Level field book for recording the staff readings, distances, and other field notes.

LEVEL

The instrument which is used for leveling, is known as level. It consist essentially of the following parts.

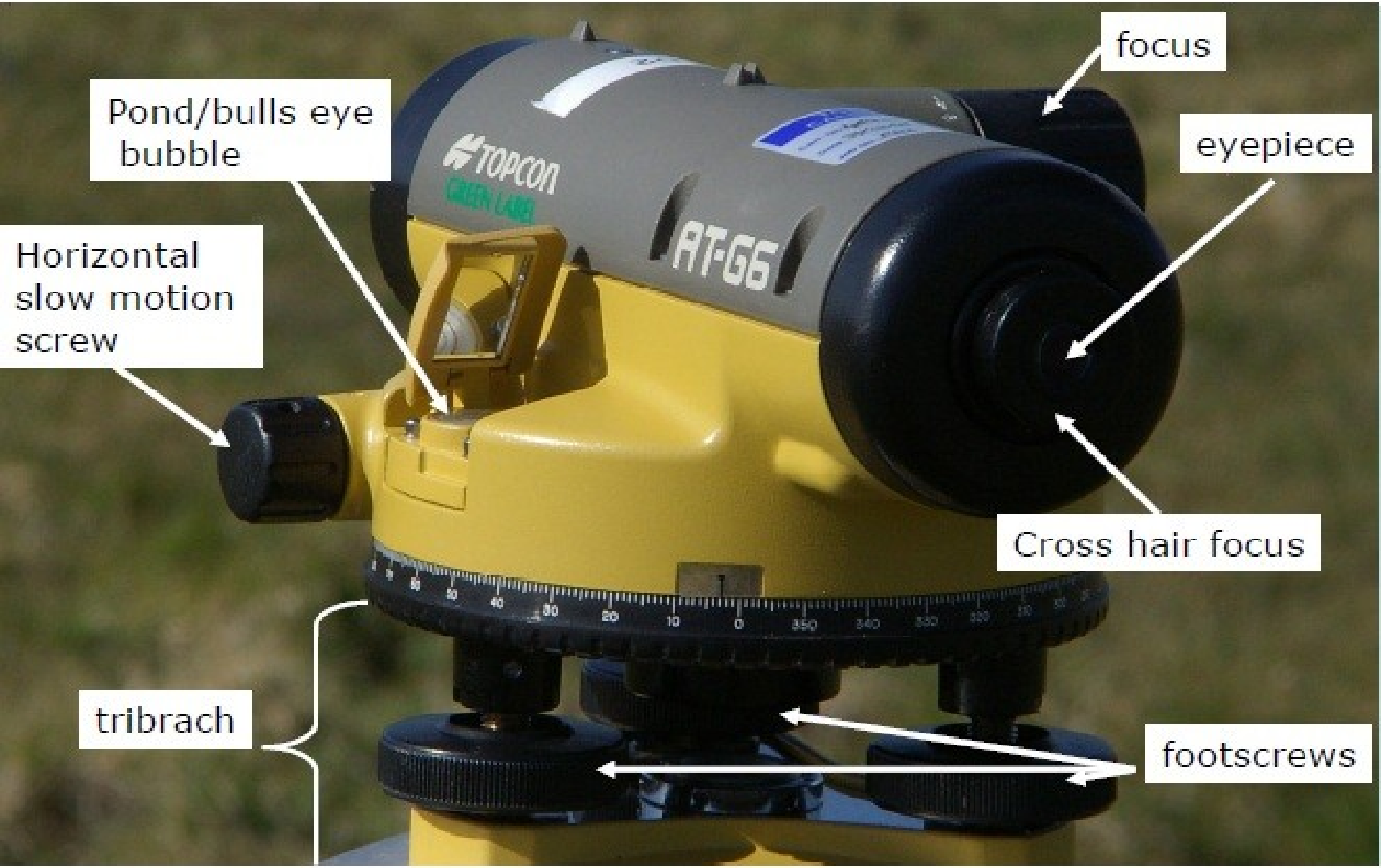
1. A telescope to provide a line of sight.

2.A level tube to make the line of sight horizontal

3.Leveling screws to bring the bubble of the tube level at the centre of its run.

4.A tripod to support the above three parts of the level.

LEVEL – BASIC COMPONENTS



WYE OR Y – LEVEL



TILTING LEVEL



AUTOMATIC LEVEL



ELECTRONIC (0R DIGITAL) LEVEL



THE LEVELING STAFF

The leveling Staff is used to determine the vertical distances of the points below the horizontal line of sight.

There are various types of graduated Staff :

- **Folding type 4-meter leveling Staff**
- **Invar precision leveling Staff**
- **Target Staff**
- **Bar Coded Staff**

For work of ordinary accuracy, we use folding staves of 4 meter length or 16 ft length. For high degree of accuracy 3 meter long invar staves are used which do not fold.

THE FOLDING STAFF



THE TARGET STAFF



BAR CODED STAFF





HOLDING THE STAFF

Utmost care should be taken in holding the staff truly vertical while the reading is taken. To hold the staff in a vertical position, the staff man stands behind the staff, heels together, with the heel of the staff between his toes, and holds it between the palm of his hands at the height of his face. If it is not vertical, the reading will be too great. Special care must be exercised with larger readings, since the error due to a given deviation from the vertical vary with the readings. In precise leveling, the staff is equipped with a folding circular level or a pendulum plumb bob to make it plumb while in ordinary leveling, the staff is waved slowly forward, i.e. towards the level, and backward, i.e. away from the level, and the lowest reading taken to avoid these errors.

HOLDING THE STAFF

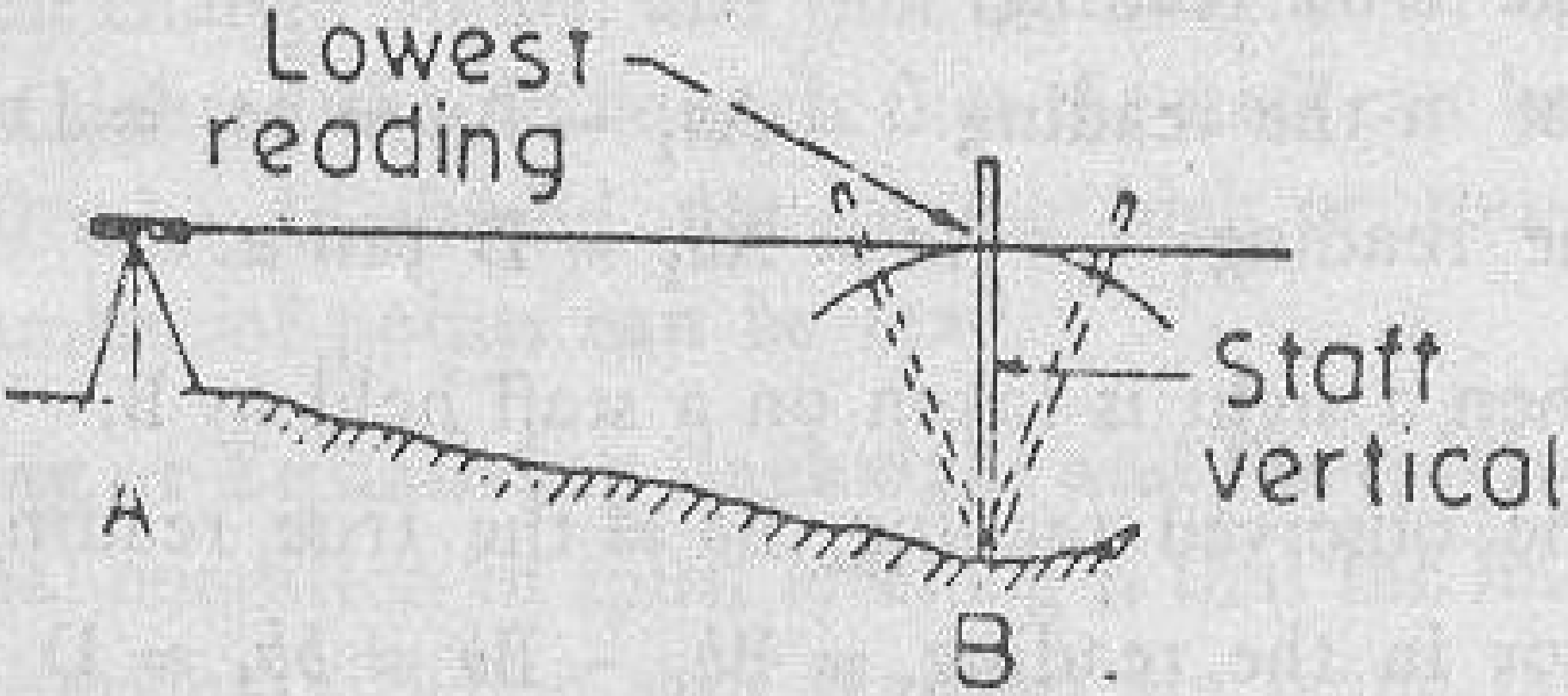


Fig. 253

BENCH MARKS

Bench Marks:

A relatively permanent and fixed reference point of known elevation above the assumed datum, is called a bench mark.

There are four kinds of bench marks:

(i) G. T. S. (Great Trigonometrical Survey)

(ii) Permanent

(iii) Arbitrary

(iv) Temporary

G.T.S Bench Marks:

These bench marks are established by the Survey of Pakistan with greatest precision at an interval of 100 Km (normally) all over all country. Their elevation refer to the mean sea level datum obtained by hourly observations of the tides over a specific time frame. These are depicted on the topo sheets published by the Survey of Pakistan and their elevations correct to two decimal points of a decimal of a meter, are entered.

Permanent Bench Marks:

These bench marks are established between G.T.S bench marks by the Survey of Pakistan , or other government agencies such as PWD, on clearly defined and permanent natural or cultural detail points such as isolated rocks culverts, kilometer stones, railway platforms, gate pillars of inspections houses, etc.

Arbitrary Bench Marks:

These are the reference points whose elevations are arbitrarily assumed for small leveling operations. Their elevations do not refer to any fixed datum as in case of G.T.S. or permanent bench marks.

Temporary Bench Marks:

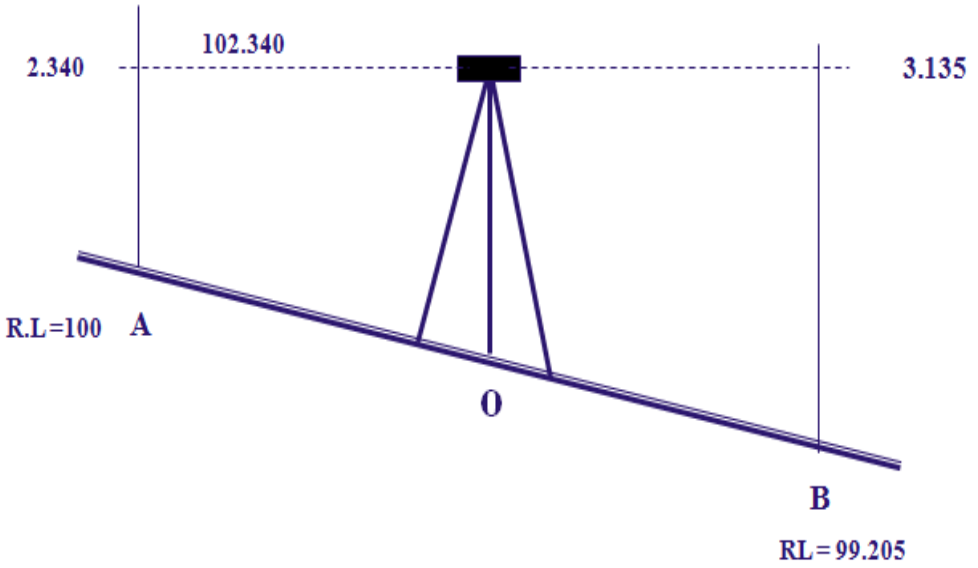
These are the reference points on which a day's work is closed and from where leveling is continued next day in the absence of a permanent bench mark. Their elevations are referred to as the reduced levels. Such bench marks are carefully established on permanent detail points such as kilometer stones, parapets, floor of verandahs, root of old trees, etc. Their correct descriptions should invariably be written in level books.

STEPS IN LEVELING

- There are two essential steps in leveling. There first step is to find the elevation or R.L. of the plane of collimation (H.I) of the level by taking a back sight on a bench mark.
- The second step is to find the elevation or R.L. of any other point by taking a reading on the staff held at that point.

PRINCIPLES OF LEVELING – SIMPLE LEVELING

It is essentially nothing but to find out the difference in the elevation (or RL) between two points, both of which are visible from a single point. Suppose A and B are two such points and the level is set up at O, approximately midway between A and B. The staff reading is then taken. Let the respective readings on A and B be 2.340m and 3.135m. The difference between these readings gives the difference of the level between A and B, which is equal to 3.135 – 2.340 = 0.795m



If the R.L of A is 100, the R.L of B may be found as:

H.I at O = 100 +2.340 = 102.340

R.L of B = 102.340 – 3.135 = 99.205

DIFFERENTIAL LEVELING

This method is used in order to find the difference in elevation between two point. (i) If they are too far apart, or (ii) If the difference in elevation between them is too great, or (iii) If there are obstacles intervening. In such a case, it is necessary to set up the level in several positions and to work in a series of stages. The basic leveling procedure is employed in each of the successive stages. The process is also known as compound or continuous leveling.

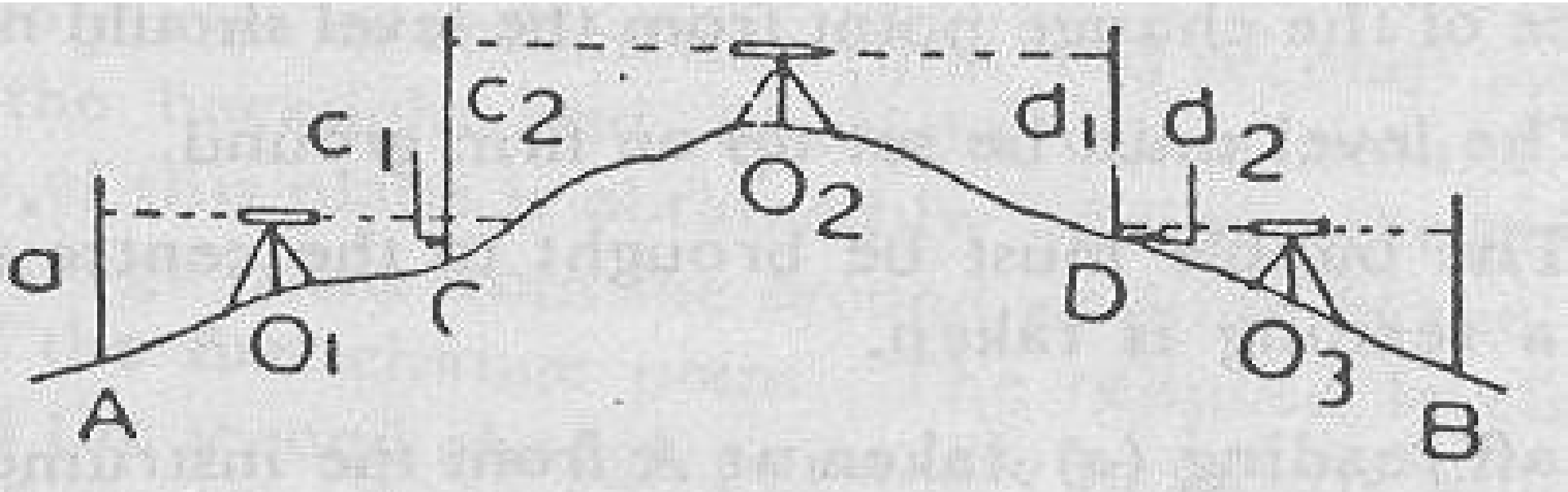
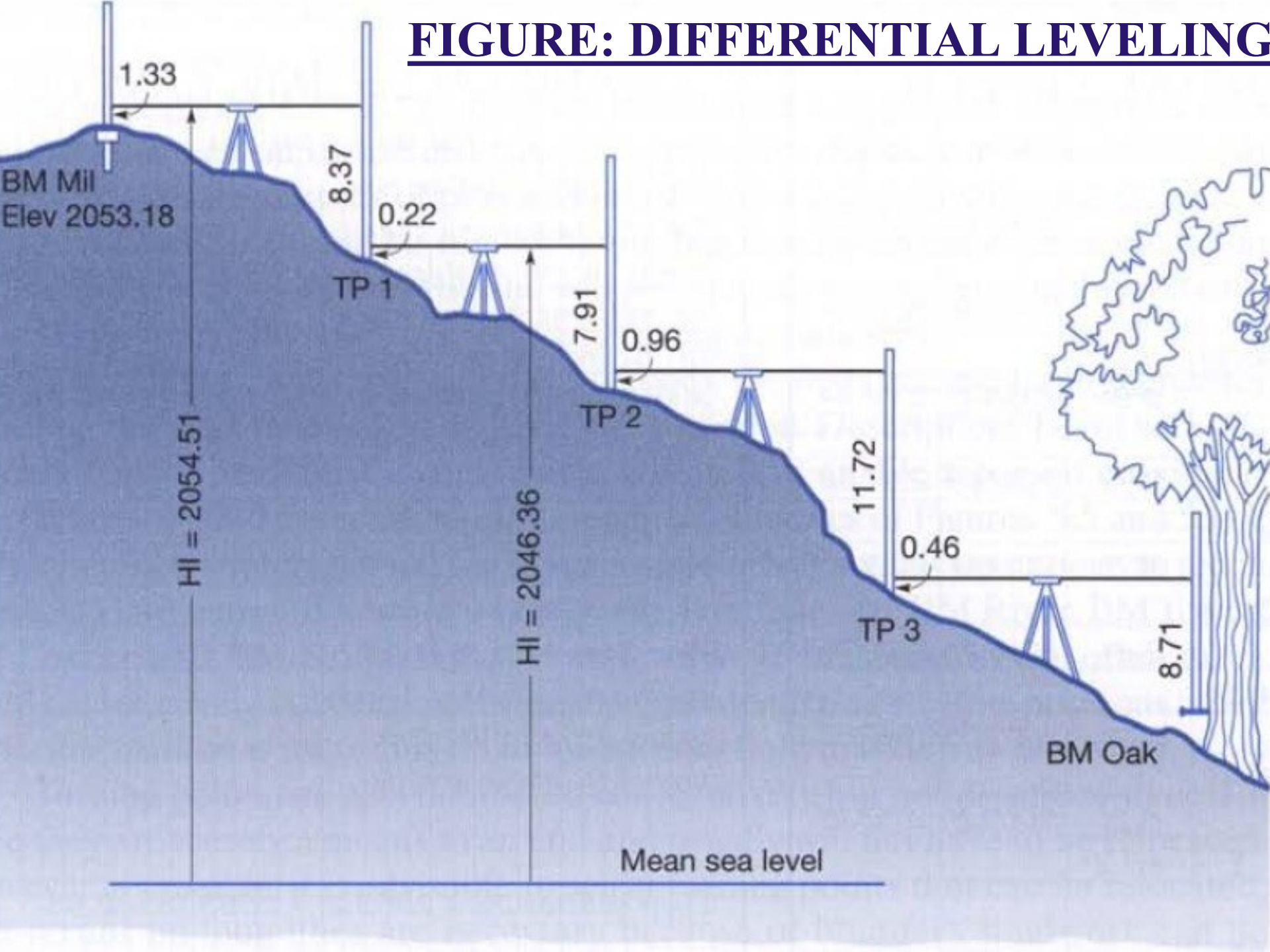


Fig. 257

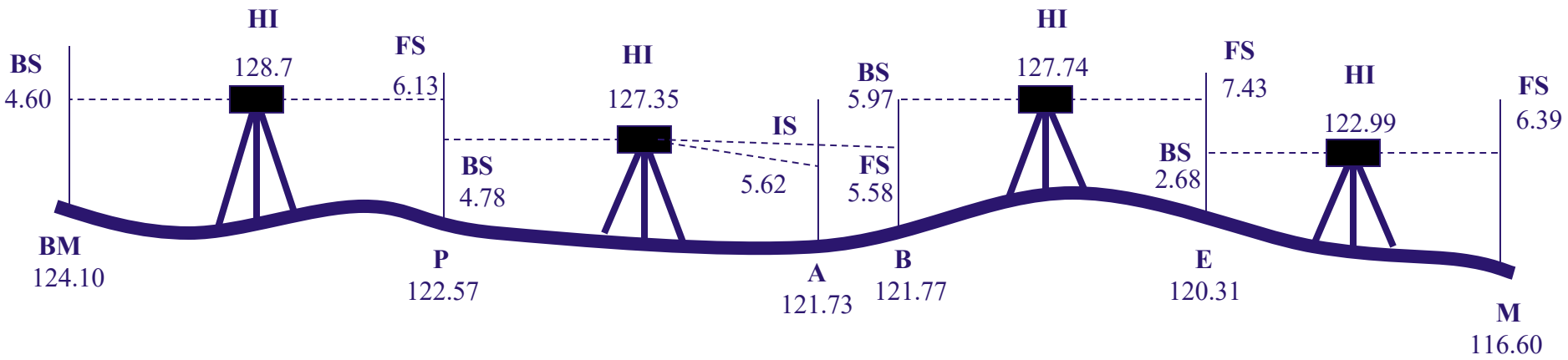
FIGURE: DIFFERENTIAL LEVELING



HEIGHT OF COLLIMATION METHOD

Station	Readings			H.I.	Reduced Level	Remarks
	Back sight	Inter sight	Fore sight			
Arith-metical Check	ΣB.S=___		ΣF.S=___	Last R.L – First R.L		
	ΣB.S-ΣF.S					

DIFFERENTIAL LEVELING – EXAMPLE



Station	Readings			Collimation	Reduced Level	Remarks
	Back sight	Inter sight	Fore sight			
1	4.60			128.7	124.10	BM
P	4.78		6.13	127.35	122.57	C.P
A		5.62		127.35	121.73	
B	5.97		5.58	127.74	121.77	C.P
E	2.68		7.43	122.99	120.31	C.P
M			6.39	122.99	116.61	

BS= Back Sight IS= Intermediate Sight

FS= Fore Sight BM= Bench Mark

HI = Height of Instrument

LEVELING BOOKING FORM

Station	Readings			Collimation	Reduced Level	Remarks
	Back sight	Inter sight	Fore sight			
1	4.60			128.7	124.10	BM
P	4.78		6.13	127.35	122.57	C.P
A		5.62		127.35	121.73	
B	5.97		5.58	127.74	121.77	C.P
E	2.68		7.43	122.99	120.31	C.P
M			6.39	122.99	116.61	
Arithmetical Check	∑BS = 18.03		∑FS = 25.53		-7.5	
	-7.5					

REDUCTION OF LEVELS

Booking and reduction of the levels may be done by following two methods.

- Height of collimation method
- Rise and fall method

REDUCTION OF LEVELS

The Height of Collimation (H.I.) Method:

It consists in finding the elevation of the plane of collimation (H.I. for every set-up of the instrument, and then obtaining the reduced levels of points with reference to the respective plane of collimation.

Arithmetical Check:- The difference between the sum of the backsights and the sum of the foresights should be equal to the difference between the first and the last reduced levels. This check verifies the calculation of the reduced levels of the planes of collimation and of the change points only. There is no check on the reduction of the intermediate reduced levels.

The Rise and Fall Method:

It consists in determining the difference of level between the consecutive points by comparing each point after the first with that immediately preceding it.

The difference between their staff readings indicates a rise or fall according as the staff reading at the point is smaller or greater than that at the preceding point. The reduced level of each point is then found by adding the rise to, or subtracting the fall from the reduced level of the preceding point.

***Arithmetical check*:-** There are three checks on the accuracy of the reductions. The difference between the sum of the back sights and the sum of the foresights = the difference between the sum of the rises and that of falls = the difference between the first and the last reduced levels.

ΣB.S – ΣF.S = ΣRISE – ΣFALL = LAST R.L – FIRST R.L

GRADIENT.

Gradient = Horizontal Distance / Difference in elevation

Summary of the Two Methods

When establishing the heights of new TBMs and other important points, only BS and FS Readings should be taken and the rise and fall method of calculation should be used.

The height of collimation method can be much quicker when a lot of intermediate sights have been taken and it is good method to use when mapping or setting out where many readings are often taken from a single instrument position.

A disadvantage of the height of collimation method is that the check on reduced levels calculated from intermediate sight readings can be lengthy and there is a tendency for it to be omitted.

HEIGHT OF COLLIMATION METHOD

Station	Readings			H.I.	Reduced Level	Remarks
	Back sight	Inter sight	Fore sight			
Arith-metical Check	ΣB.S=___		ΣF.S=___	Last R.L – First R.L		
	ΣB.S-ΣF.S					

HEIGHT OF COLLIMATION METHOD

Station	Readings			H.I.	Reduced Level	Remarks
	Back sight	Inter sight	Fore sight			
1.	2.191				49.873	T.B.M. 1
2.		2.505				A
3.		2.325				B
4.	3.019		1.496			C (C.P.)
5.		2.513				D
6.	1.752		2.811			E (C.P.)
7.			3.824			T.B.M. 2
Arith-metical Check	ΣB.S=__		ΣF.S=___	Last R.L – First R.L		
	ΣB.S-ΣF.S					

HEIGHT OF COLLIMATION METHOD

Station	Readings			H.I.	Reduced Level	Remarks
	Back sight	Inter sight	Fore sight			
1.	2.191			52.064	49.873	T.B.M. 1
2.		2.505			49.559	A
3.		2.325			49.739	B
4.	3.019		1.496	53.587	50.568	C (C.P.)
5.		2.513			52.074	D
6.	1.752		2.811	52.528	50.776	E (C.P.)
7.			3.824		48.704	T.B.M. 2
Arith-metical Check	ΣB.S = 6.962		ΣF.S = 8.131	Last R.L = 48.704 First R.L = 49.873		
	ΣB.S-ΣF.S=-1.169			Last R.L – First R.L= - 1.169		

ADJUSTMENTS OF LEVELS – COLLIMATION METHOD

Table 2.2 • Height of collimation method.

BS	IS	FS	HPC	Initial RL	Adj	Adj RL	Remarks
2.191			52.064	49.873		49.873	TBM 1 49.873
	2.505			49.559	+0.002	49.561	A
	2.325			49.739	+0.002	49.741	B
3.019		1.496	53.587	50.568	+0.002	50.570	C (CP)
	2.513			51.074	+0.004	51.078	D
1.752		2.811	52.528	50.776	+0.004	50.780	E (CP)
		3.824		48.704	+0.006	48.710	TBM 2 48.710
Checks							
6.962		8.131		48.704			
8.131				49.873			
–1.169				–1.169			

Misclosure = 48.704 – 48.710 = – 0.006 m = – 6 mm
Hence, correction is positive!!!

RISE & FALL METHOD

Station	Readings			Rise	Fall	Reduced Level	Remarks
	Back sight	Inter sight	Fore sight				
1							
2.							
3.							
4.							
5.							
6.							
7.							
8.							
9.							
Arith-Metical Check	∑B.S-∑F.S = ∑Rise - ∑ Fall = Last R.L – First R.L						

RISE & FALL METHOD

Station	Readings			Rise	Fall	Reduced Level	Remarks
	Back sight	Inter sight	Fore sight				
1	2.191					49.873	T.B.M.
2.		2.505					A
3.		2.325					B
4.	3.019		1.496				C (C.P.)
5.		2.513					D
6.	1.752		2.811				E (C.P.)
7.			3.824				T.B.M.
8.							
9.							
Arith-Metical Check	**∑B.S-∑F.S = ∑Rise - ∑ Fall = Last R.L – First R.L**						

RISE & FALL METHOD

Station	Readings			Rise	Fall	Reduced Level	Remarks
	Back sight	Inter sight	Fore sight				
1.	2.191					49.873	T.B.M. 1
2.		2.505			0.314	49.559	A
3.		2.325		0.180		49.739	B
4.	3.019		1.496	0.829		50.568	C (C.P.)
5.		2.513		0.506		51.074	D
6.	1.752		2.811		0.298	50.776	E (C.P.)
7.			3.824		2.072	48.704	T.B.M. 2
8.							
9.							
Arith-Metical Check	**∑B.S-∑F.S = ∑Rise - ∑ Fall = Last R.L – First R.L** 6.962 – 8.131 = 1.515 – 2.684 = 48.704 – 49.873 -1.169 = -1.169 = -1.169						

ADJUSTMENTS OF LEVELS – RISE/FALL METHOD

Table 2.1 • Rise and fall method.

BS	IS	FS	Rise	Fall	Initial RL	Adj	Adj RL	Remarks
2.191					49.873		49.873	TBM 1 49.873
	2.505			0.314	49.559	+0.002	49.561	A
	2.325		0.180		49.739	+0.002	49.741	B
3.019		1.496	0.829		50.568	+0.002	50.570	C (CP)
	2.513		0.506		51.074	+0.004	51.078	D
1.752		2.811		0.298	50.776	+0.004	50.780	E (CP)
		3.824		2.072	48.704	+0.006	48.710	TBM 2 48.710
Checks								
6.962		8.131	1.515	2.684	48.704			
8.131			2.684		49.873			
–1.169			–1.169		–1.169			

Misclosure = 48.704 – 48.710 = – 0.006 m = – 6 mm
Hence, correction is positive!!!

HOW GOOD IS OUR LEVELING???

On construction sites and other engineering projects, leveling is usually carried out over short distances and it can include a lot of instrument positions. For this type of work:

- **Allowable Misclosure = ± m√n**

where:

'm' is a constant based on site conditions ; usually its value is taken as **5** mm. For setting out earthwork excavations, the value of m can be raised to **30** mm. For setting out steel & concrete structures, the value of m can be lowered to **3** mm

&

'n' is the number of instrument positions.

In our example problem; allowable misclosure comes out to be ± 5√3 = ± 9 mm. Hence our misclosure is acceptable.

SOURCES OF ERRORS IN LEVELING

❑ Instrumental Errors

- Line of sight (or collimation) not truly horizontal
- Cross hair not exactly horizontal
- Staff not correct length
- Tripod legs loose

❑ Natural Errors

- Curvature of the Earth
- Refraction
- Temperature Variations
- Winds
- Settlement of the instrument/change point

SOURCES OF ERRORS IN LEVELING

❑ Personal Errors

- Bubble not centered
- Parallax
- Faulty/incorrect staff reading(s)
- Improper staff handling
- Settlement of the instrument/change point

LEVELLING EXERCISE